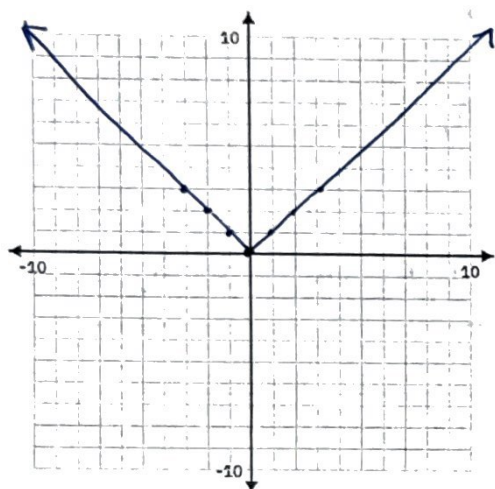


NOTES 2A.1- Graphing Absolute Value Functions

In this unit you will learn to:

- graph absolute value transformations
- give a verbal description of the transformations
- write equations from descriptions and graphs
- List all relevant information about a transformed graph
- Write a function notation description

Notes:



Parent Function Absolute Value

$$f(x) = |x|$$

table

X	X
-3	3
-2	2
-1	1
0	0
1	1
2	2
3	3

$$D: (-\infty, \infty) \quad R: [0, \infty)$$

$$x\text{-int: } (0, 0) \quad y\text{-int: } (0, 0)$$

$$\text{Vertex: } (0, 0)$$

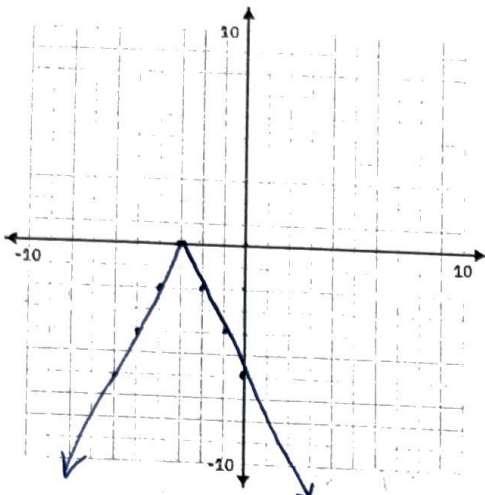
$$\text{min at } (0, 0)$$

$$\boxed{\text{axis of symmetry}} + \boxed{x=0} \text{ always } \updownarrow$$

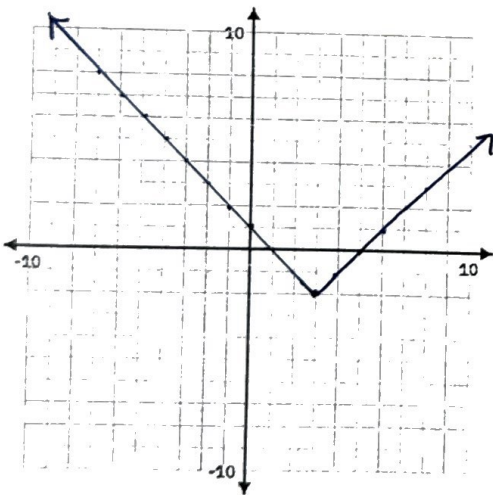
Absolute Value Function Transformations

Key Concepts	Vertical Translation			
	$f(x) + d$ $f(x) = x + d$	$f(x) + 3$ $f(x) = x + 3$	Vertical Translation Up 3	$f(x) - 3$ $f(x) = x - 3$ Vertical Translation Down 3
	Horizontal Translation			
	$f(x - c)$ $f(x) = x - c $	$f(x - 3)$ $f(x) = x - 3 $	Horizontal Translation Right 3	$f(x + 3)$ $f(x) = x + 3 $ Horizontal Translation Left 3
	Vertical Stretch and Compression			
	$af(x)$ $f(x) = a x $	$3f(x)$ $f(x) = 3 x $	Vertical stretch by a factor of 3 (stretch when $ a > 1$)	$\frac{1}{3}f(x)$ $f(x) = \frac{1}{3} x $ Vertical compression by a factor of $\frac{1}{3}$ (compression when $0 < a < 1$)
	Horizontal Stretch and Compression			
	$f(bx)$ $f(x) = bx $	$f(\frac{1}{3}x)$ $f(x) = \frac{1}{3}x $	Horizontal stretch by a factor of 3 (stretch when $0 < b < 1$)	$f(3x)$ $f(x) = 3x $ Horizontal compression by a factor of $\frac{1}{3}$ (compression when $ b > 1$)
	Reflection			
		$-f(x)$ $f(x) = - x $	Reflection about the x-axis	$f(-x)$ $f(x) = -x $ Reflection about the y-axis

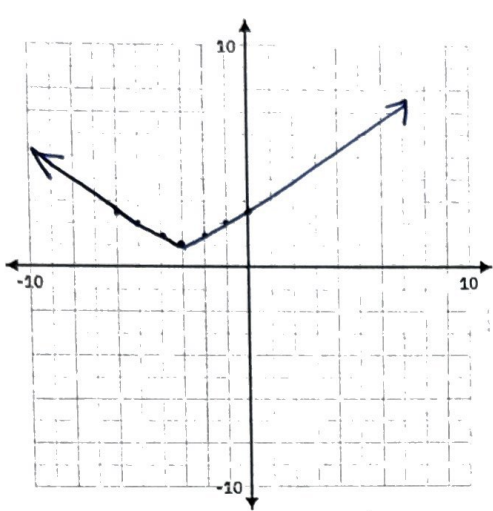
Example 1: Given an equation, graph, and provide all information.

Equation: $f(x) = -2 x + 3 $	Domain: $(-\infty, \infty)$
	Range: $(-\infty, 0]$
	X-int: $(-3, 0)$
	Y-int: $(0, -6)$
	a o s : $X = -3$
	Vertex: $(-3, 0)$
Verbal Description: Reflect over x-axis, Vert. Str. by factor of 2, left 3	Max/Min: $(-3, 0)$
Function Notation Description: $-2 f(x+3)$	

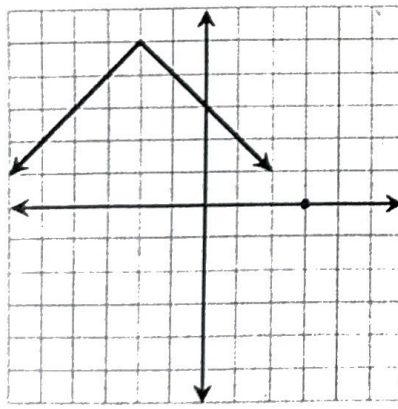
Example 2: Given an equation, graph, and provide all information.

Equation: $f(x) = x - 3 - 2$	Domain: $(-\infty, \infty)$
	Range: $[-2, \infty)$
	X-int: $(1, 0)$ $(5, 0)$
	Y-int: $(0, -1)$
	a o s : $X = 3$
	Vertex: $(3, -2)$
Verbal Description: Right 3, Down 2	Max/Min: $(3, -2)$
Function Notation Description: $f(x-3) - 2$	

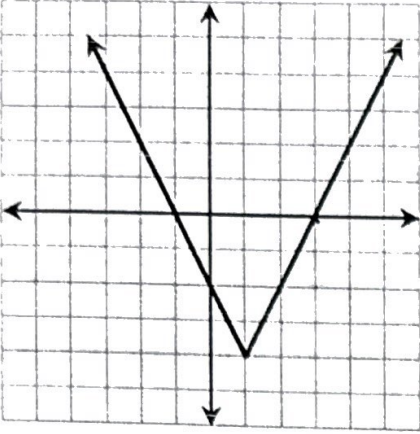
Example 3: Given an equation, graph, and provide all information.

Equation: $f(x) = \frac{1}{2} x + 3 + 1$	Domain: $(-\infty, \infty)$
	Range: $[1, \infty)$
	X-int: none
	Y-int: $(0, 2.5)$
	a.o.s: $x = -3$
	Vertex: $(-3, 1)$
	Max/Min: $(-3, 1)$
Verbal Description: Vertical Comp. by a factor of $\frac{1}{2}$, Left 3, Up 1	
Function Notation Description: $\frac{1}{2}f(x+3) + 1$	

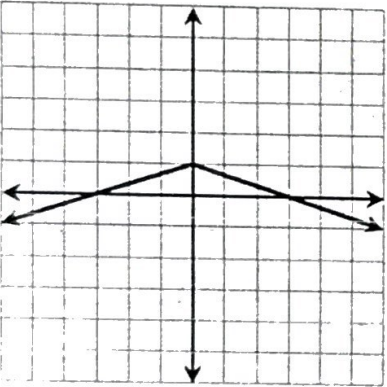
Example 4: Given the transformed graph, write the equation, and provide all information.

Equation: $f(x) = - x+2 + 5$	Domain: $(-\infty, \infty)$
	Range: $(-\infty, 5]$
	X-int: $(3, 0)$ $(-7, 0)$ (if you could see them)
	Y-int: $(0, 3)$
	a.o.s: $x = -2$
	Vertex: $(-2, 5)$
	Max/Min: $(-2, 5)$
Verbal Description: Reflect x-axis, Left 2, Up 5	
Function Notation Description: $-f(x+2) + 5$	

Example 5: Given the transformed graph, write the equation, and provide all information.

Equation: $f(x) = 2 x-1 - 4$	Domain: $(-\infty, \infty)$
 instead of "over 1, up 1" now, it's "over 1, up 2"	Range: $[-4, \infty)$
	X-int: $(-1, 0) (3, 0)$
	Y-int: $(0, -2)$
	a o s: $x=1$
	Vertex: $(1, -4)$
Verbal Description: Right 1, Down 4, Vertical stretch by factor of 2	
Function Notation Description: $2f(x-1) - 4$	

Example 6: Given the transformed graph, write the equation, and provide all information.

Equation: $f(x) = -\frac{1}{3} x + 1$	Domain: $(-\infty, \infty)$
	Range: $(-\infty, 1]$
	X-int: $(-3, 0) (3, 0)$
	Y-int: $(0, 1)$
	a o s: $x=0$
	Vertex: $(0, 1)$
	Max/Min: $(0, 1)$
Verbal Description: Reflect x-axis, Up 1, Vertical compression by factor of $\frac{1}{3}$	
Function Notation Description: $-\frac{1}{3}f(x) + 1$	

Example 7: Given the verbal description, write the equation, graph, and provide all information.

Equation: $f(x) = -\frac{1}{3}|x-4|$

Domain: $(-\infty, \infty)$

Range: $(-\infty, 0]$

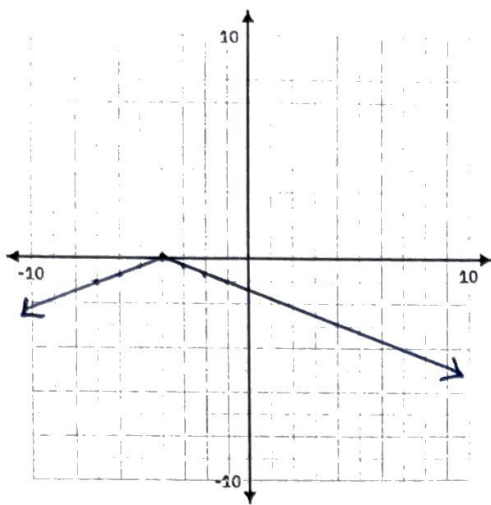
X-int: $(-4, 0)$

Y-int: $(0, -\frac{4}{3})$

AOS: $x = -4$

Vertex: $(-4, 0)$

Max/Min: $(-4, 0)$



Verbal Description: The function is reflected about the x-axis, vertically compressed, by a factor of $\frac{1}{3}$ and then shifted left 4 units.

Function Notation Description: $-\frac{1}{3}f(x-4)$

Example 8: Given the verbal description, write the equation, graph, and provide all information.

Equation: $f(x) = |\frac{1}{2}x| - 10$

Domain: $(-\infty, \infty)$

Range: $[-10, \infty)$

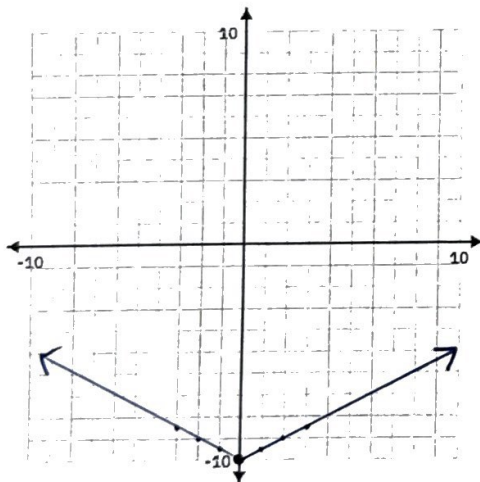
X-int: the graph isn't large enough for accuracy

Y-int: $(0, -10)$

AOS: $x = 0$

Vertex: $(0, -10)$

Max/Min: $(0, -10)$



Verbal Description: The function is stretched horizontally by a factor of 2, then translated 10 units down.

Function Notation Description: $f(\frac{1}{2}x) - 10$